Dynamic causal effects III

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Plan for this slide deck

1. Using VARs for causal inference
2. Application: monetary policy shocks
3. VAR vs. LP estimators
4. Other identification approaches

Structural VAR

- Assume we have an $n \times 1$ data vector $\boldsymbol{y}_t$ whose dynamics are well described by a VAR (here 1 lag for simplicity):

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_{t-1} + \boldsymbol{u}_t, \quad \boldsymbol{u}_t \sim WN(\boldsymbol{\Sigma}). \tag{$\dagger$}$$

- The elements of $\boldsymbol{u}_t$ are forecast errors, *not* structural shocks.
- But let's now assume that the forecast errors are linear combinations of some underlying, unobserved structural shocks:

$$\boldsymbol{u}_t = \boldsymbol{H}\boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim WN(\boldsymbol{I}). \tag{$\ddagger$}$$

- It's conventional to assume that $\boldsymbol{\varepsilon}_t$ is also $n \times 1$. Then $\boldsymbol{H}$ is $n \times n$, and $\boldsymbol{I}$ is the $n \times n$ identity matrix.
  - Note: now we normalize $\text{Var}(\varepsilon_{jt}) = 1$ for all $j = 1, \ldots, n$.
- The model ($\dagger$)–($\ddagger$) is called a **structural VAR** (SVAR).

IRFs

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_{t-1} + \boldsymbol{H}\boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim WN(\boldsymbol{I})$$

- Exercise: the (stationary) SVAR model implies the SVMA representation

$$\boldsymbol{y}_t = \boldsymbol{\mu} + \sum_{\ell=0}^{\infty} \boldsymbol{\Theta}_\ell \boldsymbol{\varepsilon}_{t-\ell}, \quad \text{where} \quad \boldsymbol{\mu} \equiv (\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{c}, \quad \boldsymbol{\Theta}_\ell \equiv \boldsymbol{A}^\ell \boldsymbol{H}.$$

  (Each of the variables $y_{1t}, \ldots, y_{nt}$ satisfies an SVMA model.)

- Impulse response at horizon h of the i-th variable $\{y_{it}\}$ wrt. the j-th shock $\{\varepsilon_{jt}\}$: (i,j) element of $\boxed{\boldsymbol{\Theta}_h = \boldsymbol{A}^h \boldsymbol{H}}$.
- In particular, the *impact* impulse responses ($h = 0$) are just $\boldsymbol{\Theta}_0 = \boldsymbol{H}$.

FEVD

- h-step-ahead forecast error:

$$\boldsymbol{y}_{t+h} - \boldsymbol{y}_{t+h|t} = \sum_{\ell=0}^{h-1} \boldsymbol{A}^{\ell} \boldsymbol{u}_{t+h-\ell} = \sum_{\ell=0}^{h-1} \boldsymbol{A}^{\ell} \boldsymbol{H} \boldsymbol{\varepsilon}_{t+h-\ell}.$$

- Let $\boldsymbol{e}_i$ be the i-th column of the $n \times n$ identity matrix $\boldsymbol{I}$.

- Fraction of forecast error variance of $y_{i,t+h}$ contributed by the sequence $\varepsilon_{j,t+1}, \ldots, \varepsilon_{j,t+h}$ of the j-th shock:

$$FEVD_{h,ij} \equiv \frac{\text{Var}\left(\boldsymbol{e}_i' \sum_{\ell=0}^{h-1} \boldsymbol{A}^{\ell} \boldsymbol{H} \boldsymbol{e}_j \varepsilon_{j,t+h-\ell}\right)}{\text{Var}\left(\boldsymbol{e}_i' \sum_{\ell=0}^{h-1} \boldsymbol{A}^{\ell} \boldsymbol{H} \boldsymbol{\varepsilon}_{t+h-\ell}\right)} = \boxed{\frac{\sum_{\ell=0}^{h-1} (\boldsymbol{e}_i' \boldsymbol{A}^{\ell} \boldsymbol{H} \boldsymbol{e}_j)^2}{\sum_{\ell=0}^{h-1} \boldsymbol{e}_i' \boldsymbol{A}^{\ell} \boldsymbol{\Sigma} (\boldsymbol{A}^{\ell})' \boldsymbol{e}_i}},$$

  where $\boldsymbol{\Sigma} \equiv \text{Var}(\boldsymbol{u}_t) = \boldsymbol{H}\boldsymbol{H}'$.

  - Exercise: verify the final expression above for the FEVD.

SVAR identification

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_{t-1} + \boldsymbol{u}_t, \quad \boldsymbol{u}_t = \boldsymbol{H}\varepsilon_t, \quad \varepsilon_t \sim WN(\boldsymbol{I})$$

- Since $\{\boldsymbol{u}_t\}$ is white noise, we learned previously that we can consistently estimate $\boldsymbol{c}$ and $\boldsymbol{A}$ with the OLS estimators $\hat{\boldsymbol{c}}$ and $\hat{\boldsymbol{A}}$.
- The final ingredient for estimating IRFs and FEVDs is $\boldsymbol{H}$. How to estimate it?
- As on the previous slide,

$$\boldsymbol{\Sigma} \equiv \text{Var}(\boldsymbol{u}_t) = \boldsymbol{H}\boldsymbol{H}'. \qquad (*)$$

- Recall that the sample var-cov matrix of the residuals, $\hat{\boldsymbol{\Sigma}}$, is a consistent estimator of $\boldsymbol{\Sigma}$.
- Given the estimate of $\boldsymbol{\Sigma}$, can we use equation $(*)$ to solve for $\boldsymbol{H}$?

SVAR identification (cont.)

$$\boldsymbol{\Sigma} = \boldsymbol{H}\boldsymbol{H}'$$

- This matrix equation has n^2 unknowns: the elements of $\boldsymbol{H}$.
- But there are only $\frac{n(n+1)}{2}$ equations, which is less than n^2 (if $n > 1$).
- Why are there not n^2 equations? Because $\boldsymbol{\Sigma}$ is necessarily symmetric, so the equations above the diagonal are redundant given the equations below the diagonal.
- Conclusion: we don't have enough equations to pin down $\boldsymbol{H}$. The IRFs are **not identified**.

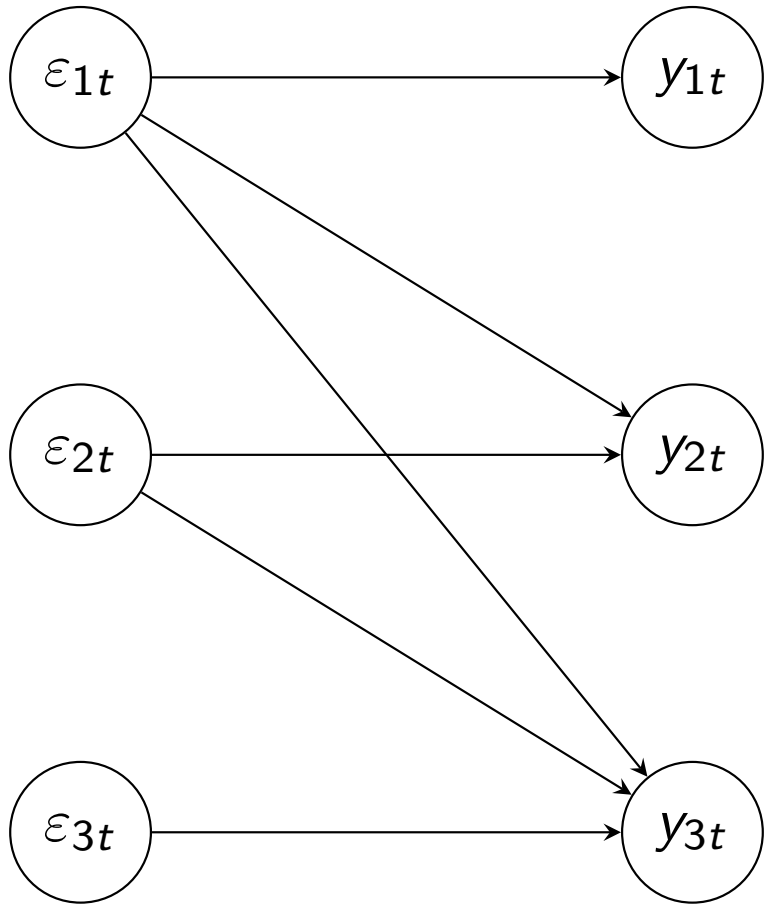
Identification from timing restrictions

- We need to make more assumptions if we want to identify $\boldsymbol{H}$.
- As in the first slide deck on local projections, we may be willing to assume some **short-run timing restrictions**: restrict which variables y_{it} can respond to which shocks ε_{jt} on impact.
- Consider "recursive" timing assumptions:

$$y_{it} \quad \text{does not respond to} \quad \varepsilon_{i+1,t}, \varepsilon_{i+2,t}, \ldots, \varepsilon_{n,t} \quad \text{on impact,}$$

  for $i = 1, 2, \ldots, n-1$.
- The last variable y_{nt} can respond to all shocks on impact.

Example with $n = 3$:



Identification from timing restrictions (cont.)

- Mathematically, these timing assumptions mean that the impact impulse response matrix $\boldsymbol{\Theta}_0 = \boldsymbol{H}$ is **lower triangular**:

$$\boldsymbol{H} = \begin{pmatrix} H_{11} & 0 & \cdots & \cdots & 0 \\ H_{21} & H_{22} & 0 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ H_{n-1,1} & H_{n-1,2} & \cdots & H_{n-1,n-1} & 0 \\ H_{n1} & H_{n2} & \cdots & H_{n,n-1} & H_{nn} \end{pmatrix}.$$

- Also assume the diagonal is positive: $H_{ii} > 0$ for $i = 1, \ldots, n$.
  - This just normalizes the sign of the shock so that y_{it} responds positively to ε_{it} on impact.

Identification from timing restrictions (cont.)

- Result (from linear algebra): the matrix equation

$$\boldsymbol{\Sigma} = \boldsymbol{H}\boldsymbol{H}'$$

  has a *unique* lower triangular solution $\boldsymbol{H}$ with positive diagonal.

- The solution is called the **Cholesky factor** of $\boldsymbol{\Sigma}$, and efficient numerical algorithms exist.
  - Python: `numpy.linalg.cholesky`
- So we can estimate $\hat{\boldsymbol{H}} \equiv \text{Cholesky}(\hat{\boldsymbol{\Sigma}})$.
- Identification from short-run timing restrictions is also called "Cholesky" or "recursive" identification.

Summary: SVAR estimation of IRF and FEVD

1. Estimate $\hat{\boldsymbol{A}}$ and $\hat{\boldsymbol{\Sigma}}$ by OLS (same as earlier slides on forecasting).
2. Compute $\hat{\boldsymbol{H}} \equiv \text{Cholesky}(\hat{\boldsymbol{\Sigma}})$.
3. Impulse response estimator:

$$\hat{\boldsymbol{\Theta}}_h \equiv \hat{\boldsymbol{A}}^h \hat{\boldsymbol{H}}.$$

   - If VAR is in $(np) \times (np)$ companion form, only apply Cholesky to $n \times n$ matrix $\hat{\boldsymbol{\Sigma}}$ and then pad $\hat{\boldsymbol{H}}$ below with zeros.

4. FEVD estimator:

$$\widehat{FEVD}_{h,ij} \equiv \frac{\sum_{\ell=0}^{h-1} (\boldsymbol{e}_i' \hat{\boldsymbol{A}}^\ell \hat{\boldsymbol{H}} \boldsymbol{e}_j)^2}{\sum_{\ell=0}^{h-1} \boldsymbol{e}_i' \hat{\boldsymbol{A}}^\ell \hat{\boldsymbol{\Sigma}} (\hat{\boldsymbol{A}}^\ell)' \boldsymbol{e}_i}.$$

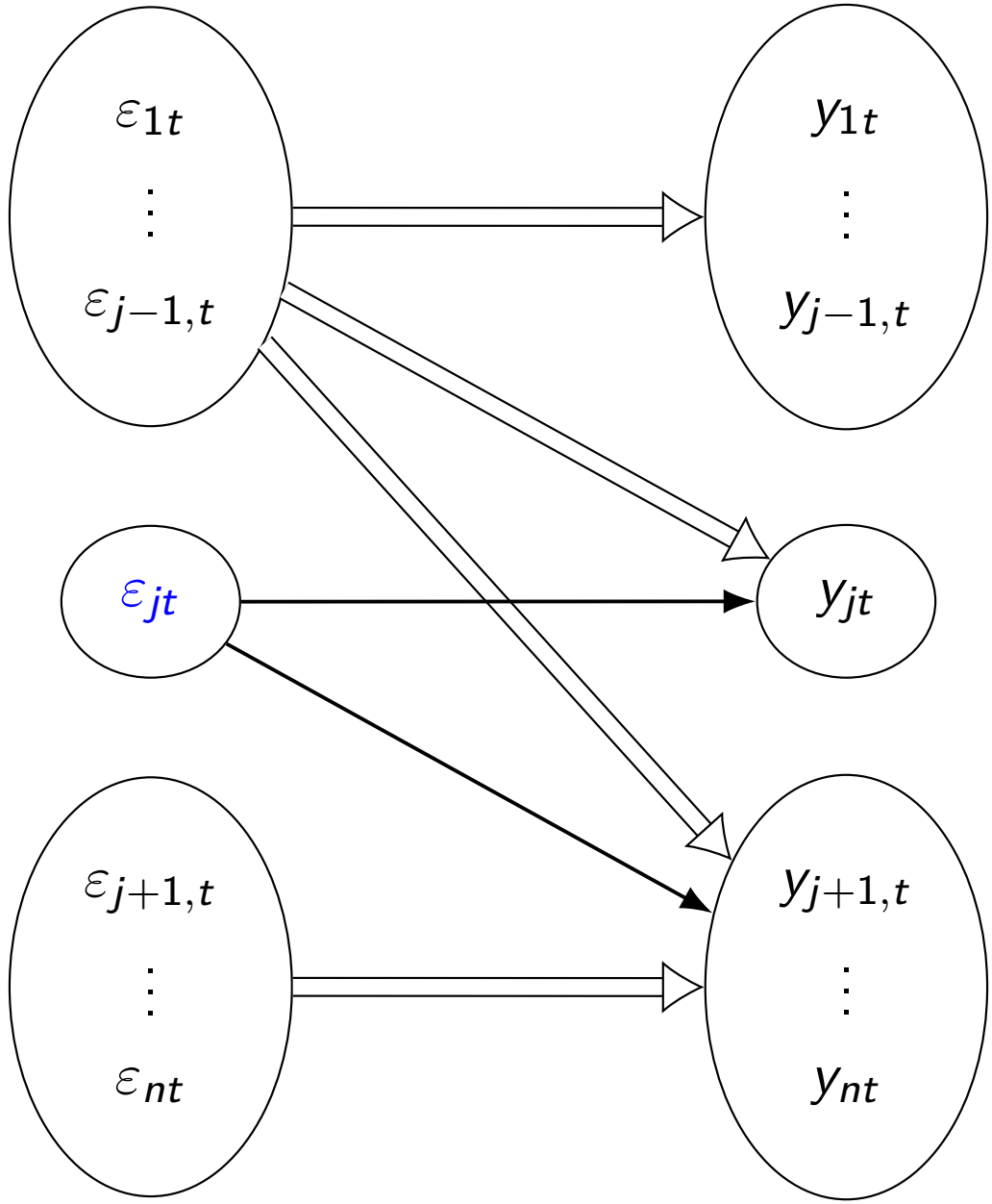
- These estimators are just functions of $(\hat{\boldsymbol{A}}, \hat{\boldsymbol{\Sigma}})$. So we can compute bootstrap confidence intervals as in our earlier slide deck on VARs.

Single-shock identification

- For the Cholesky decomposition, the ordering of the variables in $\boldsymbol{y}_t$ matters. The ordering encodes our assumptions about which shocks can affect each variable.
- If you want to identify the IRFs wrt. all the n shocks in $\boldsymbol{\varepsilon}_t$, you have to justify your specific variable ordering by defending many timing restrictions ($\frac{n(n-1)}{2}$ of them).
- But suppose you only care about a single shock, say, the j-th one.
- Then it turns out that for the Cholesky decomposition, the ordering *among* the earlier variables $y_{1t}, \ldots, y_{j-1,t}$ does not matter, and neither does the ordering *among* the later variables $y_{j+1,t}, \ldots, y_{nt}$.

Single-shock identification (cont.)

- As a consequence, when estimating the IRFs and FEVDs for a single shock j, we "only" need to assume:
  1. The first $j-1$ variables $y_{1t}, \ldots, y_{j-1,t}$ do not respond to the last $n-j+1$ shocks $\varepsilon_{jt}, \ldots, \varepsilon_{nt}$ on impact.
  2. The j-th variable y_{jt} does not respond to the last $n-j$ shocks $\varepsilon_{j+1,t}, \ldots, \varepsilon_{nt}$ on impact.
- In practice, it is often difficult enough to defend these timing assumptions for single-shock identification.
- Trying to identify all n shocks via recursive timing restrictions is rarely credible, though many papers try...


ε_{1t}
$\vdots$
$\varepsilon_{j-1,t}$
y_{1t}
$\vdots$
$y_{j-1,t}$
ε_{jt}
y_{jt}
$\varepsilon_{j+1,t}$
$\vdots$
ε_{nt}
$y_{j+1,t}$
$\vdots$
y_{nt}

Application: identification of monetary policy shocks

- We follow a classic approach to identify the causal effects of monetary policy (MP) shocks: movements in the federal funds rate beyond the systematic policy rule.
- Recursive SVAR identification with the following ordering ($n = 6$):

$$\boldsymbol{y}_t = \begin{pmatrix} \log(\text{INDPRO}_t) \\ \text{UNRATE}_t \\ \log(\text{CPI}_t) \\ \log(\text{PCOM}_t) \\ \text{FEDFUNDS}_t \\ \text{NFCI}_t \end{pmatrix}.$$

- INDPRO: industrial production. UNRATE: unemployment rate. PCOM: commodity price index. NFCI: financial conditions index.

Application: monetary policy shocks (cont.)

- FEDFUNDS ordered 5th. MP shock $\equiv \varepsilon_{5t}$.
- "Slow-moving" variables ordered *before* FEDFUNDS: INDPRO, UNRATE, CPI, PCOM.
- "Fast-moving" variable ordered *after* FEDFUNDS: NFCI.
- Timing assumptions for identifying MP shock (certainly debatable!):
  1. Slow-moving variables do not respond on impact to the MP shock ε_{5t} or to the financial shock ε_{6t}. Could be justified theoretically based on economic adjustment costs and nominal rigidities in prices.
  2. FEDFUNDS (the MP instrument) does not react to the financial shock ε_{6t} on impact. Could be justified by the Fed claiming that they "see through" short-term movements in financial markets.

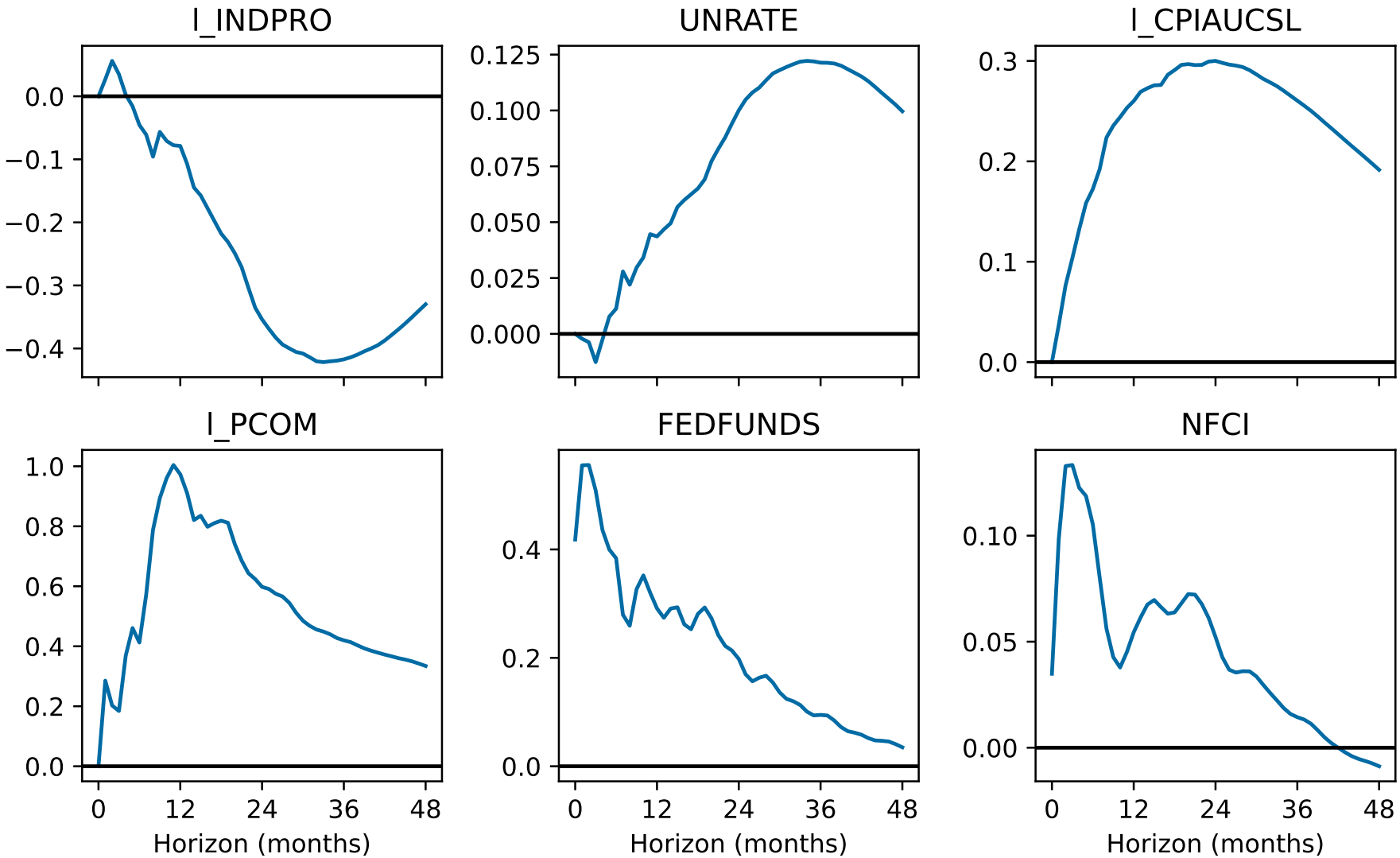
```
# Estimate VAR
order = ['l_INDPRO', 'UNRATE', 'l_CPIAUCSL', 'l_PCOM', 'FEDFUNDS', 'NFCI']
nlag = 12
model = sm.tsa.VAR(df[order])
res = model.fit(maxlags=nlag, ic=None) # fix lag length

# Cholesky-identified IRFs
max_horizon = 48
irf = res.irf(max_horizon)

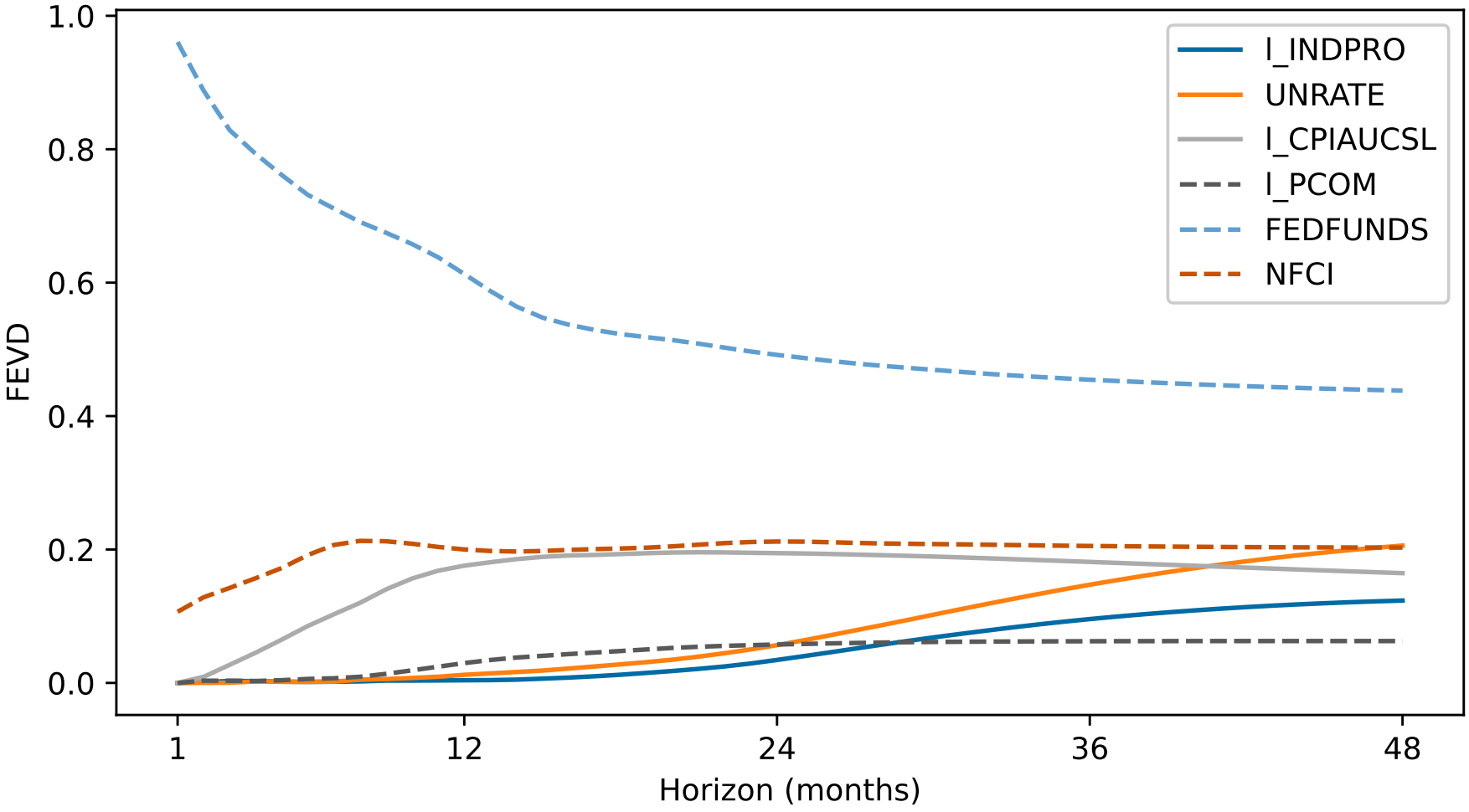
# FEVD
fevd = res.fevd(max_horizon)
```

- The structural impulse response at horizon h of variable i wrt. shock j is stored in `irf.orth_irfs[h,i-1,j-1]`.
  - "Orth": orthogonalized. The structural shocks ε_{jt}, $j = 1, \ldots, n$, are orthogonal to each other (i.e., mutually uncorrelated).
- The FEVD is stored in `fevd.decomp[i-1, h-1, j-1]`.

IRFs to 1-standard-deviation monetary policy shock



FEVD share of monetary policy shock



Trending variables

- In the application my VAR included variables with clear time trends, like industrial production.
- While our theory assumed stationary $\boldsymbol{y}_t$, it turns out that one can still compute IRFs and FEVDs using our previous formula even if $\boldsymbol{A}$ does not have all eigenvalues inside the unit circle.
- In most cases, we can also use our normal VAR bootstrap even when the variables have stochastic trends.
- We will discuss non-stationary models explicitly later in the course.

Directly measured shock

- Suppose we directly observe a shock $x_t = \varepsilon_{1t}$.
- Then we can order it first in a recursive SVAR:

$$\boldsymbol{y}_t = (x_t, y_{2t}, y_{3t}, \ldots, y_{nt})'.$$

- Since there are no variables ordered before x_t, the timing assumption is just: x_t does not respond to other shocks than ε_{1t} on impact. This is automatically satisfied if we have correctly measured the shock!
- So we can report Cholesky-orthogonalized IRFs and FEVDs wrt. the first shock in the SVAR.
  - The ordering of the other variables does not matter.

Proxies

- Now suppose we want to allow for measurement error. So our observed variable is a proxy

$$z_t = \gamma \varepsilon_{1t} + \xi_t \quad (+ \text{ lagged data}).$$

- The measurement error causes attenuation bias, just like for LP.
- As with LP, we can solve this by reporting unit-effect-normalized impulse responses:
  1. Estimate recursive SVAR with z_t ordered first.
  2. Compute IRFs $\hat{\Theta}_{h,i1}$, $i = 1, \ldots, n$, $h = 0, 1, \ldots, h_{\max}$.
  3. Report scaled IRF $\hat{\Theta}_{h,i1} / \hat{\Theta}_{0,\bar{i}1}$. Interpretation: impulse response of $y_{i,t+h}$ wrt. a shock ε_{1t} of a magnitude that raises $y_{\bar{i}t}$ by 1 unit on impact.
- Warning: FEVDs are biased if z_t has measurement error. No way to solve this without further assumptions.

Application: monetary policy shocks (cont.)

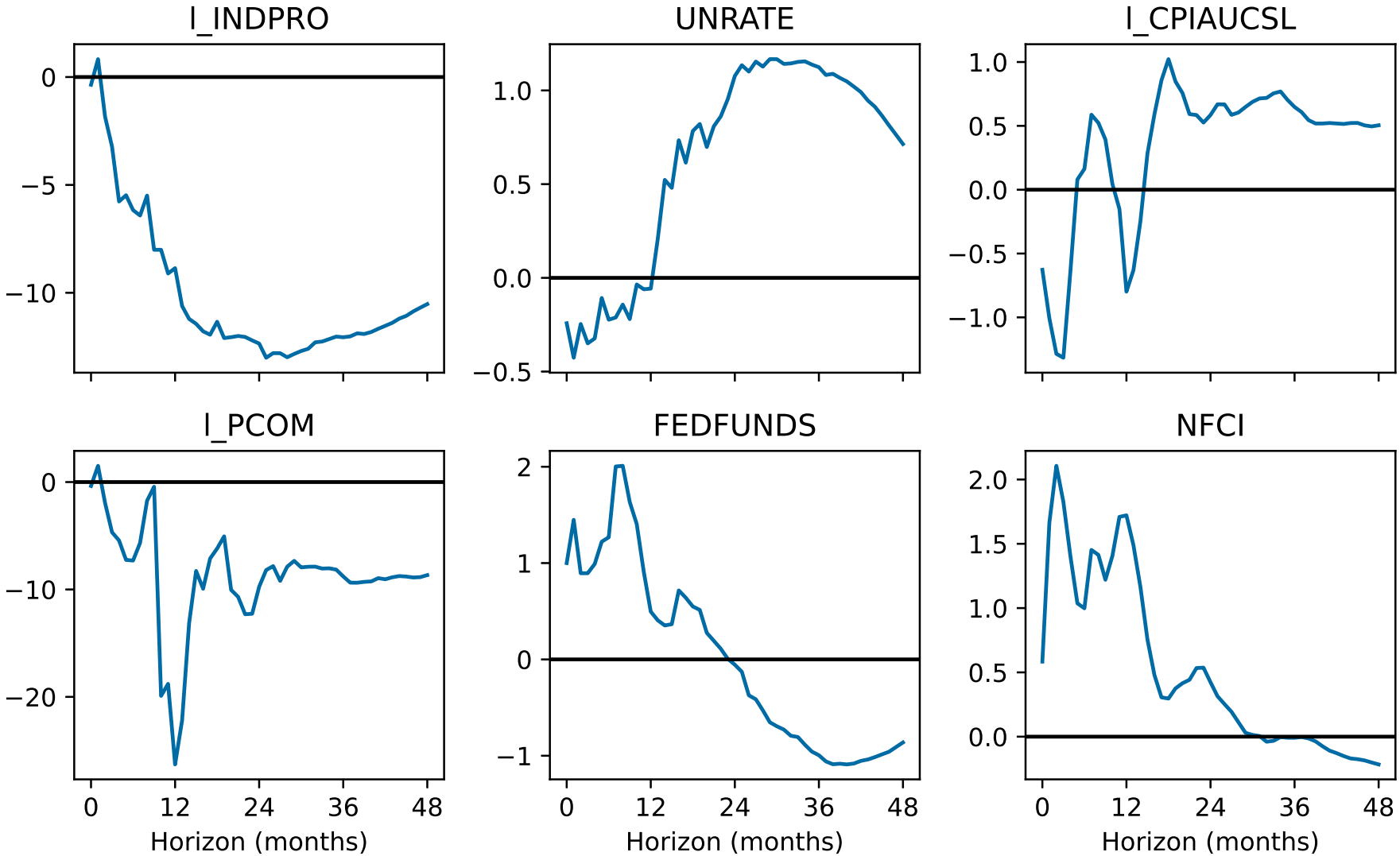
- New variable z_t: directly measured MP shock using high-frequency identification around FOMC announcements (Gertler & Karadi).
  - Updated and refined by Bauer & Swanson (2023).
- z_t is likely contaminated by financial noise.
- We order z_t first in a recursive SVAR and report unit-effect-normalized IRFs.
- Normalization: shock raises FEDFUNDS by 1 unit (100 basis points) on impact.

```
# Order shock first in SVAR
model_mps = sm.tsa.VAR(df_mps[['MPS_ORTH'] + order])
res_mps = model_mps.fit(maxlags=12, ic=None)

# Un-normalized IRFs
irf_mps = res_mps.irf(max_horizon)

# Unit effect normalization
mps_index = 0 # proxy ordered first
ffr_index_new = res_mps.names.index('FEDFUNDS')
ffr_impact = irf_mps.orth_irfs[0, ffr_index_new, mps_index]
irf_mps_normalized = irf_mps.orth_irfs[:, :, mps_index] / ffr_impact
```

IRFs to monetary policy shock (unit effect normalization)



Proxies (cont.)

- Warning: in the literature, people also use an alternative approach to proxy identification that does not directly put z_t in the estimated VAR.
- Instead, they estimate a VAR without z_t first, yielding residuals $\hat{\boldsymbol{u}}_t$. Subsequently, they regress z_t on $\hat{\boldsymbol{u}}_t$ to identify the shock of interest.
- This latter approach is sometimes called "proxy-SVAR" or "SVAR with external instrument". This approach requires more assumptions than the method that just puts z_t in the VAR and orders it first, which is sometimes called "SVAR with **internal instrument**".
- For details, see Stock & Watson (2018) on the reading list.

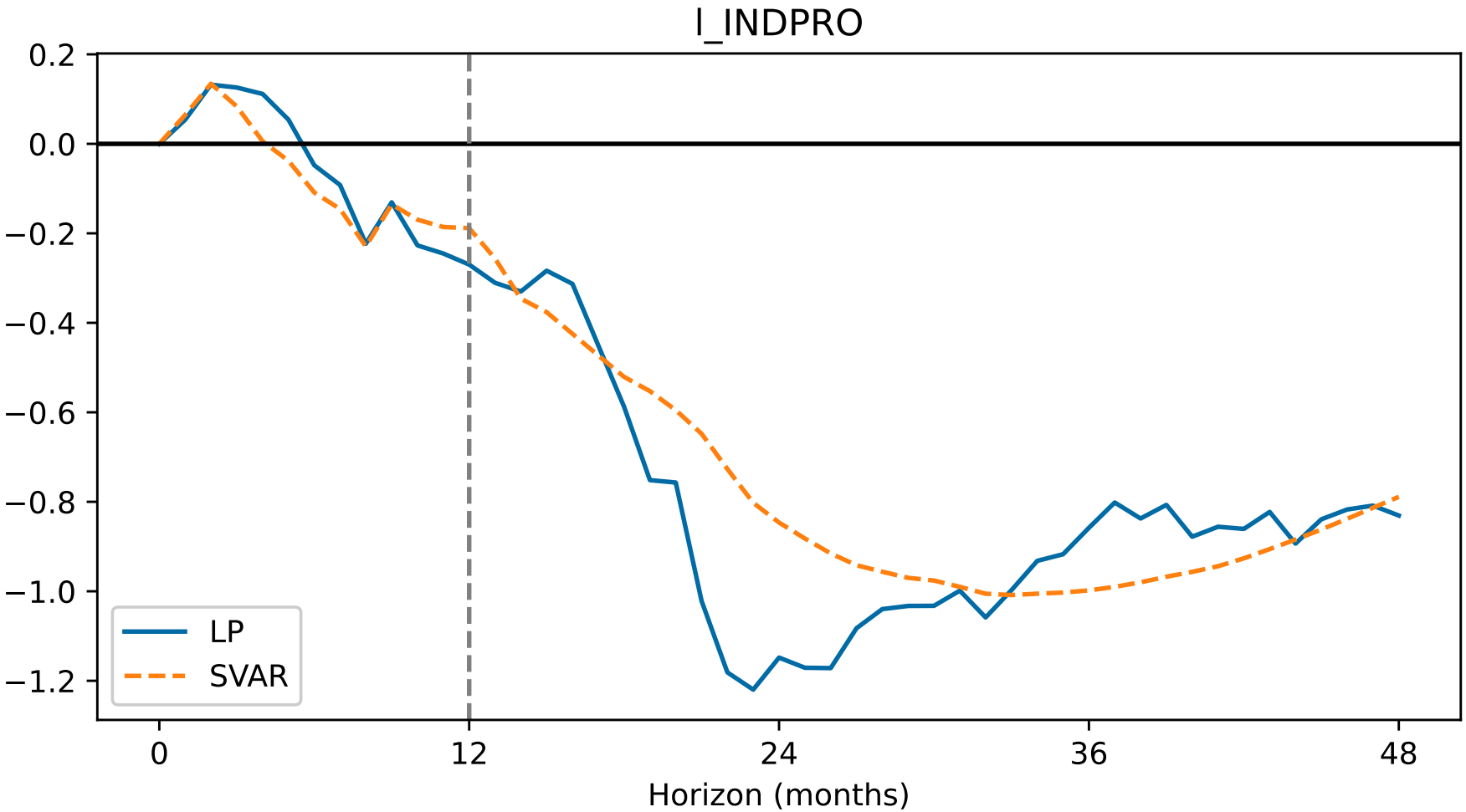
LP vs. VAR

- Both LPs and VARs can be used to estimate IRFs and FEVDs, with either observed shocks, timing restrictions, or proxies.
- Result: in large samples, the LP-estimated IRFs approximately coincide with the VAR-estimated IRFs at horizons $h \leq p$, where p is the number of lagged controls.
- At longer horizons, the two methods can differ substantially:
  - The SVAR estimate $\hat{\mathbf{\Theta}}_h = \hat{\boldsymbol{A}}^h \hat{\boldsymbol{H}}$ **extrapolates** the responses at horizons $h \geq p+1$ from the sample autocovariances at lags $1, 2, \ldots, p$.
  - The LP estimate directly uses the sample autocovariance $\widehat{\text{Cov}}(y_{i,t+h}, \tilde{x}_t)$ out to the horizon h of interest.

LP vs. VAR: recursive identification

- As we've seen, both LP and SVAR can be used for single-shock recursive identification (timing restrictions).
- Given your timing assumptions, whatever variables you would order *before* the policy variable in an SVAR should be the variables that enter as *contemporaneous* controls in an LP.
- Monetary policy shock application: LP regresses future outcomes on FEDFUNDS, controlling for...
  - *Contemporaneous* values of INDPRO, UNRATE, CPI, PCOM.
  - *Lagged* values of all six time series.

IRF of INDPRO to MP shock (unit effect normalization)



Recursive identification, normalizing impact FEDFUNDS response to 1

Vertical line marks lag length $p = 12$

LP vs. VAR: bias-variance trade-off

- There is no meaningful distinction between LP and VAR if we control for many lags p.
- For a smaller number of lags p, there is a **bias-variance trade-off**:
  - Since SVAR extrapolates, it's more exposed to bias if we get the lag length or set of controls slightly wrong.
  - For the same reason, SVAR has lower variance than LP: sample autocovariances at short lags more precisely estimated than at long lags.
- The choice of LP vs. VAR therefore comes down to your taste for bias vs. variance.
- However, when it comes to *confidence intervals*, the choice is clear: the LP interval has more accurate coverage. This is because the biased SVAR estimate can seriously degrade the coverage.

LP vs. VAR: recommendation

- Macro data sets are small, so estimation uncertainty is often large.
- We therefore want accurate uncertainty assessments, i.e., confidence intervals with accurate coverage.
- My recommendation: use LP.
  - This is just for estimating dynamic causal effects. VARs with small/moderate lag length are still very useful for forecasting!
- Alternatively, you could use an SVAR with a very large lag length (many more than what AIC typically selects). Compare your results to LP – if they are different, include more lags.
- Further discussion: Montiel Olea et al. (2026) on the reading list.

Other identification approaches

- If you read papers, you will see other ways of identifying IRFs, including (but not limited to):

1. **Long-run restrictions:** restrict which variables can respond to which shocks in the *long run* (as opposed to on impact).
   - Can be quite sensitive – difficult to robustly estimate long-run responses.
2. **Sign restrictions:** rather than assuming that certain impact responses are zero, we assume that we know the *signs* (positive/negative) of certain responses.
   - Often easier to justify based on economic theory. But inference is *very* tricky, and many papers get it wrong!

- If you want to delve into this, ask me!

What did we learn?

- SVAR
  - IRF, FEVD
  - Identification problem
  - Timing assumptions (Cholesky, recursive)
  - Directly measured shock, proxy
- LP vs. VAR
  - Similar with large p, bias-variance trade-off with small/moderate p
  - Choose LP (or VAR with large p) for accurate conf. interval coverage
- Other identification approaches

Looking ahead

- Using prior information to reduce the variance of VAR forecasts/IRFs
- Bayesian inference with Stan